

March 2026 EDA Summary

NYC TLC High-Volume For-Hire Vehicle (HVFHV) Trip Records

1. Dataset Overview

The March 2026 dataset contained **22,058,358 raw trips across 25 original columns**, making March the highest-volume month in the Q1 dataset. After removing **1 exact duplicate**, the final analytical dataset contained **22,058,357 trips**.

The two providers were **HV0003 (15.96M trips)** and **HV0005 (6.10M trips)**. Data-quality issues including negative fares, extreme distances, zero-time trips and timestamp inconsistencies were investigated and flagged rather than automatically removed.

A major timestamp finding was that **294,808 trips had request time later than pickup time**, while **no trips had pickup after drop-off**. This means the core passenger-trip sequence remained valid, and pickup/drop-off timestamps were treated as the reliable basis for trip-duration analysis.

March also contained a daylight-saving-time transition. Although **4,024,922 records did not exactly match timestamp-derived duration**, only **11,754 differed by more than one second**, with many larger differences occurring around the approximately one-hour DST shift. These observations were retained rather than incorrectly treated as invalid.

2. Key EDA Findings

Demand Patterns

March recorded a clear and consistent hourly demand pattern:

- **6 PM:** highest demand — **1,255,655 trips**
- **3 AM:** lowest demand — **323,891 trips**
- 6 PM demand was approximately **3.9×** the 3 AM level.

Demand increased through the day, reached its highest level around the evening period, and declined overnight.

This is particularly important because **January, February and March all showed the same evening demand peak around 6 PM**, making it one of the strongest recurring Q1 findings.

Trip Characteristics

March trips were predominantly short.

- **Average distance:** 4.80 miles
- **Median distance:** 2.80 miles
- **25th percentile:** 1.46 miles
- **75th percentile:** 5.94 miles
- **Average duration:** 19.24 minutes
- **Median duration:** 15.62 minutes

Both distance and duration were strongly right-skewed, meaning a relatively small number of very long trips increased the averages. The median therefore provides a better representation of a typical trip than the mean alone.

Demand vs. Trip Characteristics

The busiest period was not the period with the longest trips.

- **6 PM:** highest trip volume
- **4 AM:** longest average distance — **6.98 miles**
- **5 PM:** shortest average distance — **4.17 miles**
- **3 PM:** longest average duration — **22.56 minutes**
- **2 AM:** shortest average duration — **16.56 minutes**

This reinforces the Q1 pattern that **high demand, longer distance and longer duration occur at different times**.

3. Provider & Geographic Patterns

Provider Market Structure

Metric	HV0003	HV0005
Trips	15.96M	6.10M
Market Share	72.34%	27.66%
Avg Distance	4.91 mi	4.51 mi
Avg Duration	19.41 min	18.79 min

Avg Base Fare	\$28.08	\$26.36
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HV0003 continued to account for approximately **72% of March trips**, consistent with January and February. Its average trip distance, duration and base fare were also higher than HV0005.

HV0003 generated approximately **\$549.24M in passenger charges**, compared with **\$196.97M for HV0005**, giving HV0003 approximately **73.60% of total passenger-charge volume**. Its passenger-charge share was therefore slightly higher than its 72.34% trip share.

These differences describe provider-level trip characteristics; they do not by themselves establish differences in efficiency or profitability.

Day × Hour Demand

The March heatmap confirmed that demand patterns vary across both **day and hour**. Overnight demand remained low, morning demand increased, and evening periods consistently showed strong activity.

This suggests that analysing **day-of-week × hour** provides more useful operational information than looking at hourly demand alone.

4. Day-of-Week & Data Quality Insights

March recorded the highest raw weekday total on **Sunday (3.47M trips)** and the lowest on **Wednesday (2.74M)**. However, this comparison requires caution because March has 31 days: Sunday, Monday and Tuesday occurred five times, while the other weekdays occurred four times.

Therefore, raw weekday totals are not directly comparable without accounting for the number of occurrences. A more defensible analysis would use **average trips per occurrence of each weekday**.

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March also contained **1,773 trips above 100 miles**, with a maximum recorded distance of **1,644.54 miles**. These extreme observations were retained and flagged because they represent a very small portion of the dataset.

5. March Business Takeaway

March 2026 recorded the **highest monthly trip volume in Q1 at 22.06 million trips**. Demand was strongly concentrated around the evening, with **6 PM consistently emerging as the peak demand hour**, while early-morning trips were longer on average.

HV0003 continued to dominate the market with approximately **72% of trips**, while also showing higher average distance, duration and fare than HV0005. Most trips remained relatively short, with a small number of extreme trips creating strong right-skew in distance and duration.

The March analysis strengthened the major Q1 themes:

**Time-based Demand → Provider Structure → Trip Characteristics → Trip Economics
→ Data Quality**

Across all three months, the recurring patterns were especially clear: **evening demand peaks, longer early-morning trips, short typical trip distances, HV0003's dominant market share, and the importance of analysing trip characteristics alongside simple trip volume.**